Performance of grid-connected PV

PVGIS-5 estimates of solar electricity generation:

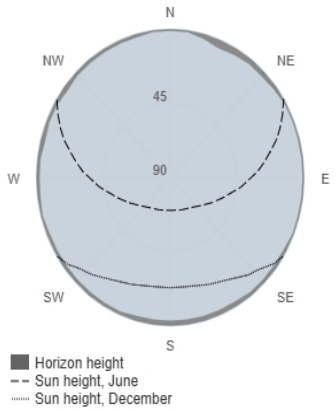
Provided inputs:

Latitude/Longitude:	42.812,-1.692
Horizon:	Calculated
Database used:	PVGIS-SARAH3
PV technology:	Cryst Sil Original
PV installed:	100 kWp
System loss:	14 %

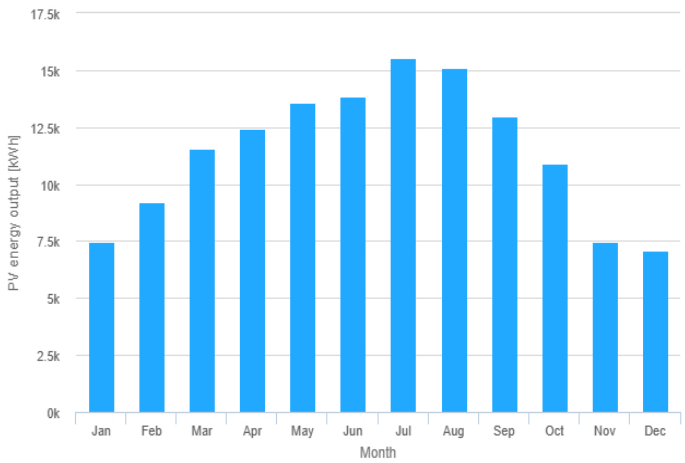
Simulation outputs

Slope angle:	35 °
Azimuth angle:	0 °
Yearly PV energy production:	137131.53 kWh
Yearly in-plane irradiation:	1794.36 kWh/m²
Year-to-year variability:	4574.42 kWh
Changes in output due to:	
Angle of incidence:	-2.75 %
Spectral effects:	1.14 %
Temperature and low irradiance:	-9.65 %
Total loss:	-23.58 %

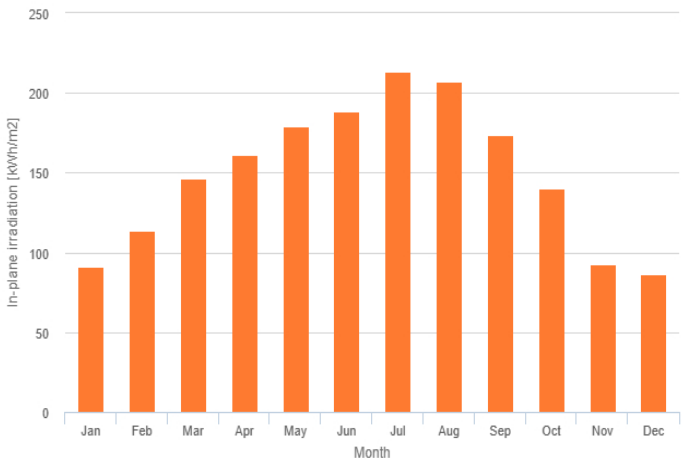
Outline of horizon at chosen location:



Monthly energy output from fix-angle PV system:



Monthly in-plane irradiation for fixed-angle:



Monthly PV energy and solar irradiation

Month	E_m	H(i)_m	SD_m
January	7465.3	90.8	1495.8
February	9199.5	113.4	1538.4
March	11554.2	146.8	1825.2
April	12410.9	161.6	1486.1
May	13595.8	179.3	1394.0
June	13851.7	188.5	931.1
July	15530.6	213.5	651.6
August	15103.3	207.2	567.4
September	12960.1	173.6	790.6
October	10878.2	140.5	905.0
November	7484.1	92.6	1242.3
December	7097.8	86.7	1037.8

E_m: Average monthly electricity production from the defined system [kWh].

H(i)_m: Average monthly sum of global irradiation per square meter received by the modules of the given system [kWh/m²].

SD_m: Standard deviation of the monthly electricity production due to year-to-year variation [kWh].

The European Commission maintains this website to enhance public access to information about its initiatives and European Union policies in general. Our goal is to keep this information timely and accurate. If errors are brought to our attention, we will try to correct them. However, the Commission accepts no responsibility or liability whatsoever with regard to the information on this site.

It is our goal to minimise disruption caused by technical errors. However, some data or information on this site may have been created or structured in files or formats that are not error-free and we cannot guarantee that our service will not be interrupted or otherwise affected by such problems. The Commission accepts no responsibility with regard to such problems incurred as a result of using this site or any linked external sites.

For more information, please visit https://ec.europa.eu/info/legal-notice_en

Joint Research Centre

PVGIS ©European Union, 2001-2026.
Reproduction is authorised, provided the source is acknowledged, save where otherwise stated.

Report generated on 2026/04/17